

Variables

Introduction to Programming

StayAhead Training; Sep, 2024

Introduction

Variables

- **Variable:** a name for some data stored in memory
- Data must exist in memory to:
 - write to an output device/disk
 - do some arithmetic
 - perform a comparison
 - make a decision

Variable Declaration and Initialisation

Variables

- Pseudocode, e.g.:
 - SET MyNumber to 42
- Pseudocode keywords are often capitalised
- Some prefer to use = instead of the word to
- Name variables as you see fit, but be consistent

Variable Declaration and Initialisation

Variables

- `my_number = 42`

Data Types

Variables

- Fundamental data types:
 - Boolean (true or false)
 - Number
 - String (text)

Data Types

Variables

- `my_boolean = True`
- `my_number = 42`
- `my_string = "Hello"`

Literals

Variables

- **Literal:** a data value
- **True** is a boolean literal
- 42 is a number (integer) literal
- "Hello" is a string literal

Variable Reassignment

Variables

- `my_boolean = False`
- `my_number = 43`
- `my_string = "Goodbye"`

Reading From Stdin

Variables

- Pseudocode, e.g.:
 - INPUT MyNumber from stdin
- There are two operations here:
 1. Input from stdin
 2. Storage of the input value in memory

Reading From Stdin

Variables

- `data = input()`

Writing to Stdout

Variables

- Pseudocode, e.g.:
 - OUTPUT MyNumber to stdout

Writing to Stdout

Variables

- `print(data)`
- `print("Hello")`